



Review Paper

Bird guilds show different responses to tree retention levels: a meta-analysis

Marco Basile^{a, *}, Grzegorz Mikusiński^{a, b}, Ilse Storch^a^a Chair of Wildlife Ecology and Management, University of Freiburg, Tennenbacher Str. 4, D-79106 Freiburg, Germany^b Swedish University of Agricultural Sciences, Department of Ecology, Grimsö Wildlife Research Station, S-730 91 Riddarhyttan, Sweden

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ABSTRACT

Since the introduction of retention forestry in the 1990s, its practice has spread in the boreal and temperate regions. Its benefits for biodiversity are well recognised, and usually presented in the form of aggregated results for each taxonomic group. However, ameliorating effects of increased levels of retention on species with different ecological requirements is still not well studied. With this meta-analysis, we aim at disentangling the response of different bird guilds, considering the level of retention applied, as well as other confounding factors, such as the forest type or harvested area. We used a systematic review protocol and collected 30 studies which compared the response of birds to a gradient of retention levels against unharvested forests and/or clearcuts. Firstly, we compared the overall response of the full bird assemblage to retention, and found no differences in abundance and species richness between retention and unharvested forest, except for the abundance of some forest specialist species. Secondly, assessing the responses of various guilds showed that each bird guild can be affected very differently, according to their life history, biome, forest type and, most importantly, retention intensity. Results allow us to infer minimum retention levels needed to preserve each bird guild, admitting that other factors can play an important role in shaping the bird assemblage, often in interaction with retention. For example, canopy and cavity nesters showed a more positive response to lower levels of retention in mixed or broadleaved forests than in conifer forests. Our results indicate that minimum retention levels ranging between 40 and 60% of original habitat are needed, in order to maintain the same bird assemblage as in the autochthonous unharvested forest, with particular regard to the forest specialists.

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1. Introduction

Today's forests in human-dominated landscapes provide valuable services, ranging from timber provision and carbon storage, to aesthetic and social values (Law et al., 2002; Schwenk et al., 2012). Timber production has been historically the main purpose of forest management, while recently, a gradual shift to multiple-use forest management integrating economic and socio-cultural values with biodiversity conservation has been observed (Bengtsson et al., 2000; Gauthier et al., 2009; Johnson et al., 1996; Paillet et al., 2010). Understanding how various components of biodiversity are affected by forestry is required for the effectiveness of such integrated approaches of forest management. Birds are a key taxonomic group of forest

* Corresponding author.

E-mail address: marcob.nat@gmail.com (M. Basile).

organisms involved in many different ecological processes (Mikusiński et al., 2018a). Forestry practices affect birds in many different ways. In general, the negative impact of logging on forest birds is well recognised (BirdLife International, 2018). Globally, logging has been found to negatively affect more than half (1383 out of 2489, 56%) of all threatened and near threatened bird species (BirdLife International, Data Zone accessed on May 22, 2017). Therefore, the relevance of birds in forest management has gained attention from both biological conservation and socio-economic perspectives (Mönkkönen et al., 2014). Specific responses of birds to forestry can be highly variable, depending on species (e.g. Twedt and Somershoe, 2009), location (e.g. Imbeau et al., 2001), spatial scale (e.g. Kellner et al., 2016), forest management regime (e.g. Vanderwel et al., 2009), and time since logging occurred (e.g. Balestrieri et al., 2018). For instance, in European coppice forests, bird assemblages turned out to be richer in younger successional stages (Deconchat and Balent, 2001; Fuller and Moreton, 1987). However, coppice forests are poor in forest-interior species and favour open-forest species, contrasting with shelterwood forests (Balestrieri et al., 2015). The latter have been shown to be beneficial for forest bird species diversity and abundance in North American forests, due to the combination of young and mature forest characteristics (King and DeGraaf, 2000), but the overall impact is still negative when compared with old-growth forests (Beese and Bryant, 1999). In recent years, retention forestry, the practice of setting aside trees or groups of trees, is gaining increasing attention as beneficial for various components of biodiversity, and especially birds, in clearcutting systems (Fedrowitz et al., 2014).

Retention forestry consists of setting aside scattered trees, tree groups or forest patches during logging operations, to retain some degree of structural variability in harvested stands, acting as 'lifeboat' and maintaining connectivity among stands (Franklin, 1989). Set-aside trees are usually existing or potential microhabitat-bearing trees, often old and decaying, also referred to as habitat trees or legacy trees (Gustafsson et al., 2012). The evidence shows that retention forestry has a generally positive impact on biodiversity and its importance will most probably increase in the future (Gustafsson et al., 2019); it has already been included in the legislation of some countries (Gustafsson et al., 2010; Soler et al., 2015).

Following the habitat heterogeneity hypothesis (MacArthur and MacArthur, 1961), retention forestry reduces the habitat simplification that usually follows intensive logging, e.g. clearcutting, and allows the forest to retain and further develop old-growth features. This process mitigates the threats to biodiversity, as predicted by the species-area-habitat hypothesis (Triantis et al., 2003), expecting higher diversity where old-growth features are preserved. Different retention levels may also mitigate habitat loss at the landscape scale (e.g. Fahrig, 2001). Andrén's (1994) review of studies on birds and mammals suggested that a major ecological change occurs when habitat cover declines to approximately 10–30% of the landscape. For instance, the difference in tree cover explained 55–60% of the variance in richness of woodland-dependent birds in Australia, with several species that declined markedly in landscapes with less than 10% tree cover (Radford et al., 2005). However, another review suggested that such critical thresholds can be highly variable and difficult to assess (Swift and Hannon, 2010).

The response of birds to retention forestry has been studied in the last 30 years, but long-term assessments are still lacking in the scientific literature (Mikusiński et al., 2018b). In 1995, Hansen et al. showed that retention forestry applied in conifer forest in Oregon (USA) can be beneficial for some bird species (4 out of 23) when compared to clearcutting, but, at the same time, 6 species showed negative responses when compared to old-growth (Hansen et al., 1995). This and other early works (Baker and Lacki, 1997; Merrill et al., 1998; Norton and Hannon, 1997) highlighted how bird response can vary depending on individual species' ecological requirements, forest ecosystem type, geographic location and the level of retention. More recently, interest arose around the effect of retention forestry on forest-specialist birds, as they can represent, among the various bird assemblages or guilds, the group most threatened by intensive forestry practices (Fuller and Robles, 2018). Here, the local retention of key structural elements (i.e. structural retention) might not be enough to ensure landscape-level persistence of forest-specialist populations (Woodley et al., 2006). Implementing retention measures across landscapes can improve the overall performance of retention (Roberge et al., 2015). As an example, retaining snags in temperate forests is an efficient practice to keep Eurasian three-toed woodpecker (*Picoides tridactylus*) populations stable (Bütler et al., 2004); it has been suggested that a minimum target of 1.3 m²/ha (basal area) of snags is sufficient. In boreal forest of eastern Canada, forests with a snag (>9 cm dbh) density > 200 snag/ha presents a higher probability of nest tree selection by the black-backed woodpecker (*Picoides arcticus*), and the authors suggest to retain at least 10 live trees and snags (>27 cm dbh) per hectare (Tremblay et al., 2015). In Southern Patagonia, Lencinas et al. (2018) proposed that applying variable retention levels was the most effective way to conserve habitat of bird species in managed forest landscapes. Forest biodiversity management should aim at preserving the complete forest bird assemblage in spite of forestry and other anthropogenic disturbances. Fedrowitz et al. (2014), in their review of the effects of retention forestry on flora and fauna, found that in retention cuts, overall bird species richness and abundance was similar to that in mature forests; however, retention was less effective in preserving richness and abundance of the forest specialist species. The mechanisms behind these patterns, however, remain obscure. Improved knowledge on the potential of retention forestry in efficiently conserving birds in managed forest landscapes is highly desirable (Mikusiński et al., 2018b; Odsen et al., 2018).

Ideally, research should propose the management schemes that attain the goal of maintaining complete bird assemblages in managed forests, based on understanding the mechanisms behind the responses of individual species to various management practices. However, taking into account the variability of species' responses to management and the complexity of forest ecosystems the pragmatic solution is to study abundance and diversity of the entire bird assemblage with focus on guilds or functional groups and not on individual species (e.g. Kroll et al., 2017).

Here, we used a meta-analysis approach aimed at disentangling the response of various bird guilds (*sensu* Blondel, 2003) to different retention levels considering the amount of retention applied, as well as potential confounding factors, such as type of forest and size of harvested area. We base our analysis at patterns in bird abundance and species richness since these two

metrics are commonly used to assess bird responses to human action, and also in monitoring schemes (Gregory et al., 2008). Finally, we also aimed at providing recommendations for retention schemes aimed at maintaining complete bird species assemblages in managed forests, and at identifying knowledge gaps and use these for delineating future research needs.

In order to tackle our objective, we firstly explored the response of entire bird assemblages and various guilds to retention forestry and, then, looked further into the response in relation to the intensity of retention. To characterize this gradient in intensity, we used three different measures: retention level (amount or number of trees retained), retention type (dispersed vs. aggregated), and retention extent (cutblock size). We considered unharvested forests as reference for our comparison, as did the studies we reviewed in this meta-analysis.

2. Methods

2.1. Literature search

We searched the ISI Web of Knowledge database using the words 'forest' AND 'bird' AND ('retention' OR 'selective harvest' OR 'partial harvest') in the topic search, for the entire available time span (as of July 2017). The search returned 340 publications, whose abstracts were screened for consistency with our general aims (e.g. studies focusing only on dead trees or on fire experiments were excluded), reducing the number to 132. Further, we defined the following criteria for a publication to be included in the meta-analysis: (1) presence of unharvested control areas; (2) abundance or species richness estimates from species surveys not focusing on pre-selected species; (3) quantification of the retention level; (4) last year of logging reported; (5) harvest area reported. After applying these criteria, the final list of articles included in our analysis encompassed 30 studies (appendix S1). From each article, we extracted the following bird data: (1) mean abundance estimate for all species, expressed as index, density or true abundance and (2) standard deviation or standard error; (3) mean species richness for the complete bird assemblage. We decided to focus on species richness instead of more complex measures of species diversity in order to be able to include more studies in our meta-analysis. In fact, while species diversity is a metric that in most cases need to be estimated by the authors of the original paper, species richness can also be derived retrospectively from the species lists provided. We then extracted the same data for the following guilds, separating each species by nest sites and foraging modes, using a common reference (Rodewald, 2015): (1) canopy nesters; (2) cavity nesters; (3) ground nesters; (4) aerial foragers; (5) bark foragers; (6) foliage foragers; (7) ground foragers. Guild data was extracted when the authors themselves estimated abundance and/or richness for these guilds or when they provided the full species inventory with single-species abundance and richness per treatment. We included all the species that nest or feed in forest. For instance, the American kestrel (*Falco sparverius*) is part of the cavity nesting guild but it does not belong to any feeding guilds, as it feeds outside of the forest. Single species' estimates of mean abundance and standard deviation were summed and square root summed, respectively. We used ImageJ (Schneider et al., 2012) to extract data from figures, through the Figure_Calibration plugin (Hessman, 2009). Finally, we extracted the following ancillary study parameters, in order to use them as additional variables of the treatment effects in the meta-analysis: (1) forest type, expressed as broadleaved, conifer or mixed forest; (2) the latitude, which was derived from toponym (i.e. using name of places), if not explicitly stated by the authors; (3) the altitude, which was derived from toponym or geographic coordinates, if not explicitly stated by the authors.

2.2. Meta-analysis

For meta-analysis (following Vetter et al., 2013), we scaled several variables to improve data handling: retention level (RL) was transformed as a ratio between basal area/ha or trees/ha or retention level (%) in treatments and unharvested areas. Therefore, the scaled variable could range between $0 \leq x \leq 1$. Many studies included clearcuts, to which we assigned a retention value of 0, if not differently expressed by the authors. The size of the harvested area (extent of the cutblock, CB) and the years after harvest (YH) were log-transformed. Altitude (AL) and latitude (LA) were normalised. Forest type (FT) was a categorical variable, including broadleaved, conifer and mixed forest. Finally, we followed the studies' descriptions of retention to separate sparse from aggregate retention, where aggregate means leaving tree groups after logging.

2.2.1. Bird response to retention vs. non-retention

Two different effect size measures were computed for abundance and richness data. We calculated the standardised mean difference (SMD) or Hedges' *g* (Hedges, 1981) from abundance data. SMD allows for contrasting the treatment against the control, taking into account the mean, the standard deviation and the sample size of both treatment and control. SMD can range between $-\infty; +\infty$ and values below or above zero indicate treatment effects higher or lower than the controls. Differently, the log incidence rate ratio (IRR) (Bagos and Nikolopoulos, 2009) was computed for the richness data. As richness data are often reported as simple species counts, without proper estimation of mean values and variability around the mean, the IRR has the advantage of requiring only counts and sample size. It can range between $-\infty; +\infty$ and values below or above zero indicate treatment effects higher or lower than the controls. We built a null model (without variables) to test the overall effect of retention against unharvested on bird assemblages and guilds, fitted by maximum likelihood. We used the same approach to distinguish between sparse retention, aggregate retention and clearcut. The null models for species abundance and species richness were tested for publication bias, using the trim and fill method (Duval and Tweedie, 2000). This method

uses a non-parametric data augmentation technique to estimate the number of studies potentially missing from the meta-analysis.

2.2.2. Bird response along a retention gradient

The two main random-effect models, species abundance and species richness, showed no or little bias in data reporting (Appendix S2), therefore we proceeded in the analysis as follows. To test the influence of retention level and the other mentioned variables on species abundance and species richness of forest birds, we employed a meta-regression approach with mixed-effect models of the response variable (i.e. SMD of abundance or IRR of richness). We assumed a normally distributed variation between the true effect sizes and the effect sizes of each article, which we included in the random term (Nakagawa et al., 2015). Consequently, we built a full model including all the mentioned variables in a multiplicative framework and the random effect. We employed a genetic algorithm to select the best variables from the candidate set, using AICc (Akaike, 1973) as selection criterion and $\Delta\text{AICc} < 2$ as inclusion threshold in the final candidate model set (Burnham and Anderson, 2002). Abundance models were fitted via restricted maximum likelihood, which can be a better choice in case of small sample size (Kenward and Roger, 1997). The richness model showed optimization problems, during the analysis, when fitted via restricted maximum likelihood, therefore we employed maximum likelihood. As final model we selected the one with the smallest number of variables (most parsimonious one) or with the variable of interest (RL) included without interaction (see details in Appendix S4 tab. S1 and S2). Our next evaluation went beyond the empirical values estimated via GLMMs. We investigated how the bird assemblage as a whole and each guild are affected along the gradient of retention, when all the other variables are kept on their average values. We therefore projected abundance and richness estimates along the retention gradient. We used a conservative approach to choose the lowest retention level. Such a level was found at the intersection of the upper confidence interval with the 0 line, along the retention gradient, which indicates no difference with unharvested forests. When the effect of retention level could not be meaningfully isolated from the model, we derived the optimal level in interaction with other variables. All the analyses were carried out in R environment (R Core Team, 2015), employing the packages 'metafor' (Viechtbauer, 2010) and 'gimulti' (Calcagno and Mazancourt, 2010).

2.2.3. Bird response to other variables

Finally, we examined patterns linking guild effect sizes and other significant variables. We used the same methodology as above, by projecting the model estimates along the variable gradients. This allowed us to further look into other forest variables, which are likely to affect birds and can be subject to management strategies.

3. Results

3.1. Data description

Among the 30 studies included in our analysis, 29 were conducted in North America, located in boreal and temperate biomes, and only one was from Europe (Slovenia) (see Appendix S1). The remaining 102 studies did not meet the inclusion criteria. The average level of retention was 0.32 ± 0.28 SD but the size of the treatment was highly variable, ranging from 1 to 841 ha (median = 14 ha). Studies lasted on average 4.0 years with only two studies lasting 10 + years. Authors carried out their studies 5.2 years, on average, after treatments, while their controls have not been harvested for $99.4 \text{ years} \pm 44.2$, on average.

Only 17 studies included abundance estimates of the entire bird assemblages, whilst 22 studies estimated bird species richness. Regarding the specific bird guilds, analysis was mainly possible when authors included the full species list in the article (as table or appendix). The collection of studies is available in the supporting information (see Appendix S3) while the number of studies included in the analysis of each guild is shown in Fig. 1.

3.2. Bird response to retention vs. unharvested

We found no difference in overall bird abundance between retention forestry and unharvested areas (Fig. 1a). This response was independent of the type of harvest employed and did not show any differences even when including clearcuts. Cavity nesters and aerial foragers were the only two guilds showing significant negative response to forestry ($\text{SMD} = -0.45 \pm 0.11$ SE; $\text{SMD} = -0.48 \pm 0.19$ SE). The negative response appeared in every harvest type, thus indicating a decrease in their abundance after treatments. Overall species richness did not show any significant response to retention forestry (Fig. 1b). Among the guilds, only cavity nesters richness was negatively affected by harvesting ($\text{IRR} = -0.31 \pm 0.15$ SE), but such response was not dependent on the harvest type.

3.3. Bird response along a retention gradient

The best model for species abundance (i.e. the model for which $\Delta\text{AICc} < 2$ and had the lowest number of variables) included altitude (AL), forest type (FT), latitude (LA), cutblock area (CB) and retention level (RL) (see Appendix S4 tab. S3). The effect of RL was found significant in interaction with CB, positively influencing the species abundance (Table 1). The effect of CB alone negatively affected species abundance, indicating a decrease in the abundance corresponding to an increase in cutblock area.

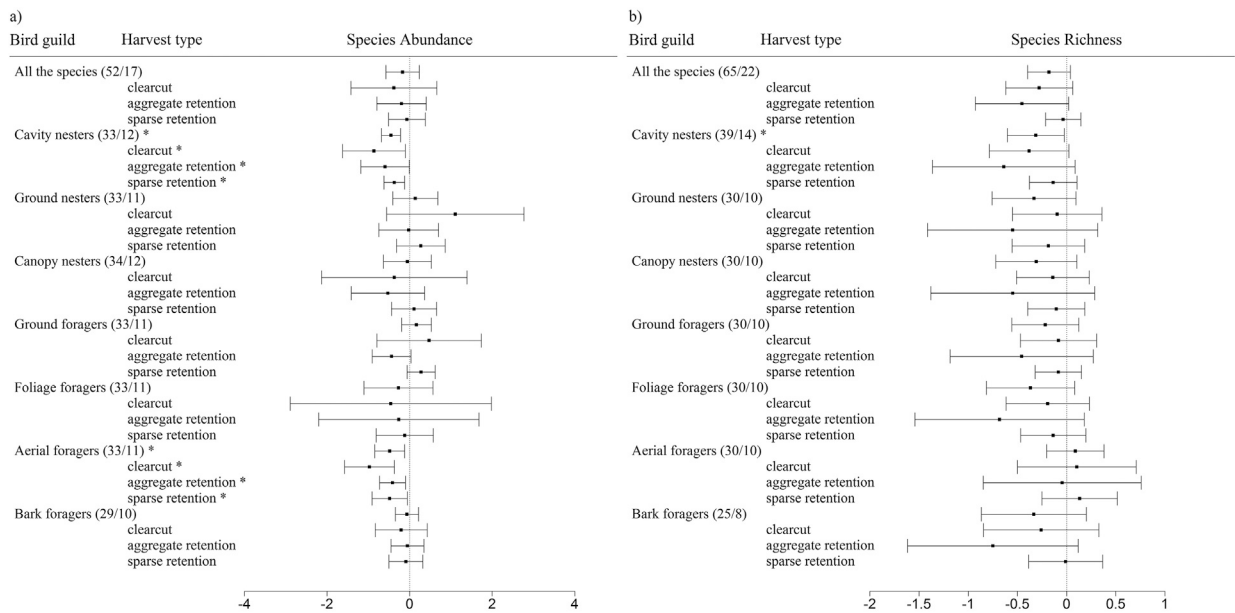


Fig. 1. Effect of retention harvests (mean $\pm$ 95% CI) on bird assemblage and guilds abundance (a) and richness (b), when compared to unharvested stands. Observed outcomes is standardised mean difference (a) and log-incidence rate ratio (b). When intervals cross 0, the effect is not significant. The first row of each group show the general effect of retention vs. unharvested, while the underlying rows show the effects grouped by retention type. Clearcuts are also shown as part of most of the included studies. In brackets are shown data points/number of studies. * denotes significant effects ($p < .05$).

Table 1

Effects of retention level on bird assemblage and guilds abundance and richness. Results are showed as positive, negative or non-significant (*n.s.*) effects and in interaction with other variables (for the magnitude and the direction of the interaction, as well as for the effects of other moderators see appendix S4). RL = retention level; AL = altitude; FT = forest type; YH = years after harvest; LA = latitude; CB = extent of the cutblock; *n.a.* = not applicable.

		Abundance		Richness	
Guilds	Acronym	RL	interaction	RL	interaction
All the species		<i>n.s.</i>	LA, CB	+	AL, LA
Canopy nesters	CAN	+	FT, LA	+	CB
Cavity nesters	CAV	+	AL	<i>n.s.</i>	LA
Ground nesters	GRN	+	AL, LA, CB	<i>n.s.</i>	<i>n.s.</i>
Aerial foragers	AEF	+	<i>n.s.</i>	<i>n.s.</i>	<i>n.s.</i>
Bark foragers	BAF	—	FT	<i>n.a.</i>	<i>n.a.</i>
Foliage foragers	FOF	<i>n.s.</i>	LA, CB	<i>n.s.</i>	<i>n.s.</i>
Ground foragers	GRF	—	YH, CB	+	<i>n.s.</i>

In addition, LA showed a significant response in interaction with RL, indicating a positive effect on species abundance with increasing latitude. Moreover, FT was also significant, showing a positive response in broadleaved and mixed forests (tab. S3). The best model for species richness included RL, as a significant variable (tab. S4). RL was significant when interacting with both altitude and latitude (tab. 1). The latter variables negatively affected species richness, indicating a decrease in the positive effect of RL at higher altitude but an increase at higher latitude. The best model for explaining canopy nesters' abundance included FT, LA and RL (tab. S3). Retention level positively affected canopy nesters' abundance, while such an effect decreased at increasing latitude (tab. 1). Forest type also played an important role, indicating a stronger response in mixed forest, (tab. S3). Canopy nesters richness was significantly affected by AL, LA, CB, RL, YH and FT (tab. S4). Specifically, RL positively affected canopy nesters' richness, but the effect decreased with increasing CB (tab. 1). The best model for cavity nesters' abundance included AL, FT, CB and RL (tab. S3). Cavity nesters' abundance increased along the retention gradient and showed similar responses among forest types (tab. 1). The latitude played an important role for cavity nesters' richness, as RL was significant when interacting with LA (tab. 1 and S2). Differently from the previous guilds, ground nesters showed negative or null response to retention level, in abundance and richness, respectively. In particular, abundance was negatively affected, although the effect reversed at increasing CB and LA, and decreasing AL (tab. 1 and S1-S2). The best model explaining aerial foragers' abundance included retention level only (tab. 1 and S1), with a significant positive effect. In contrast, aerial foragers' richness was not significantly influenced by RL (tab. 1 and S2). Bark foragers' abundance response to RL was variable among forest types (tab. 1 and S1). While an overall negative response characterised this guild, the response in mixed forest appeared strongly positive, compared to the other forest types (tab. S3). The best model for foliage foragers' abundance included FT, LA, CB, RL and YH (tab. S3). Retention level was significant in interaction with CB and LA (tab. 1), indicating that RL effect increased

with CB, while having more pronounced effect in temperate latitudes (tab. S3). Retention level did not influence significantly foliage foragers' richness, which was affected by AL, FT, LA and CB (tab. 1 and S2). Ground foragers' abundance showed a negative response to LA and RL (tab. S3). The latter became positive when in interaction with CB and YH (tab. 1 and S1). Regardless, the confidence interval always included the 0, indicating null effects are likely. Ground foragers' richness, instead, was positively affected by AL and RL, while LA affected it negatively (tab. S4).

3.4. Retention levels to preserve the bird assemblage

As described above, the entire bird assemblage and the guilds showed very different responses to retention level. We derived retention levels needed to preserve the same abundance and/or richness each guild as in unharvested forests. Species abundance did not show any significant retention level, displaying a general positive response to increasing level of retention, more pronounced in temperate forests, as well as in mixed and broadleaved (Fig. 2a). Species richness crossed the 0, i.e. the value where there is no difference between retention cut and unharvested forests, when the retention was 0.47 (Fig. 3a). Canopy nesters' responses in abundance varied depending on the forest type and being generally positive in boreal forests. On the contrary, the response in richness was homogeneous across forest types and systems, with retention cuts predicted to have richness values similar to unharvested forests at a level equal or higher than 40%, Fig. 3b. Cavity nesters' abundance confidence intervals indicated a minimum level of retention ranging 41–59%, depending on forest types. However, their response in richness was different also between boreal and temperate forests, showing a positive effect of RL in boreal forests (with a level of 60% for broadleaved forest) and a negative effect in temperate forests (Fig. 3c). Aerial foragers and bark foragers showed also generally positive responses to retention, with levels of 58% and 70% (only for conifer forests), respectively (Fig. 2 e–f). The response of foliage foragers was different in magnitude among forest types and systems, though always showing the same positive direction. The projections, then, showed that retention levels between 41 and 45% could be enough in broadleaved and conifer forests (Fig. 2g). Finally, ground foragers showed a non-significant trend in their abundance, while richness was positively affected, showing effect sizes above 0 for RL > 43% (Fig. 3d).

3.5. Response to other variables

Effect sizes showed different patterns relative to CB and YH depending on bird guilds. Tree-dependent guilds, for which YH was significant (i.e. canopy nesters, foliage foragers and bark foragers) showed a similar response to YH, with a decrease in the abundance in the years immediately after the harvest took place (see Appendix S5; figs S1, S2). On the contrary, ground related guilds showed a null, or slightly increasing response after the harvest, declining after some decades (see Appendix S5; fig. S3). Among the guilds for which CB was significant, all showed a bell-shaped response in abundance to CB, except for ground nesters, which showed a decline (see Appendix S5; fig. S4 – S6). Richness, instead, increased with patch size for both tree and ground dependent guilds (see Appendix S5; fig. S7, S8).

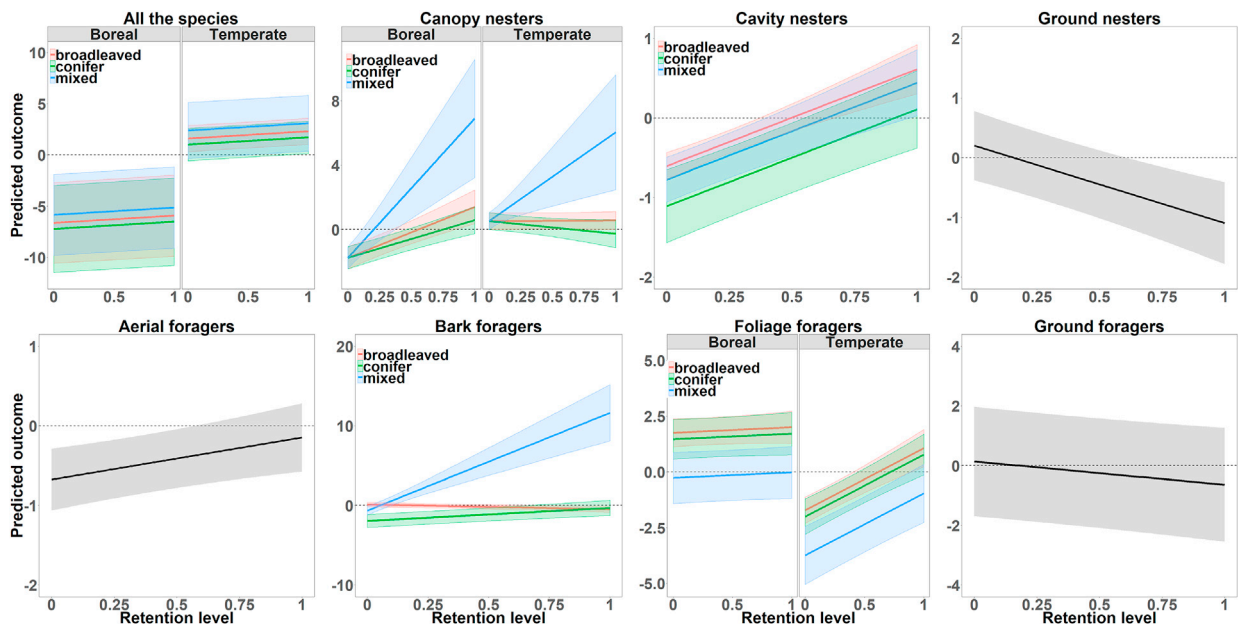


Fig. 2. Effects of retention level on the abundance of bird assemblage (a) and guilds (b–h). Only significant relationships are shown. When significant, they are also showed in interaction with forest types or forest system (by latitude, assuming boreal forests are present above the 50° parallel). The 0 line denote the absence of differences in abundance with unharvested/mature forests. Above the line the abundance in retention forests is greater than in unharvested forests.

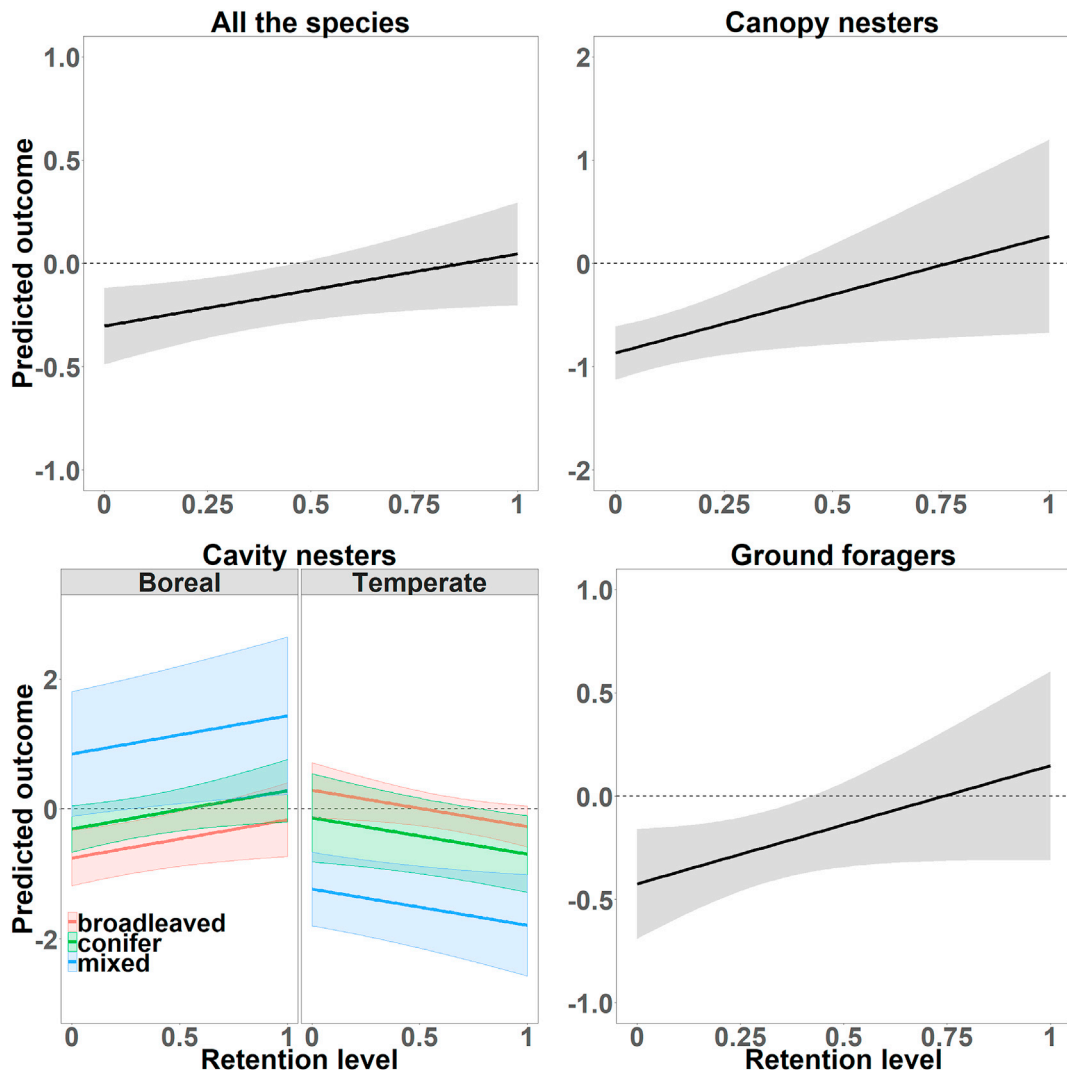


Fig. 3. Effects of retention level on the richness of bird assemblage (a) and guilds (b–h). Only significant relationships are shown. When significant, they are also showed in interaction with forest types or forest system (by latitude, assuming boreal forests are present above the 50° parallel). The 0 line denote the absence of differences in richness with unharvested/mature forests. Above the line the richness in retention forests is greater than in unharvested forests.

4. Discussion

4.1. Overview of the results

In this meta-analysis, we investigated the effects of retention forestry on bird assemblages and guilds over a gradient of retention levels, and assessed the role of some spatial features of retention, such as cutblock extent and aggregation of retained trees. Our overall results were consistent with those of previous work (Fedrowitz et al., 2014; Mori et al., 2017): the meta-analysis showed evidence of a negative response of forest specialists to harvesting regardless of retention levels, but no differences in overall species abundance between retention and unharvested forests.

We remark that, even if our analysis can provide an extensive overview of the numerical responses of the bird assemblages to retention levels (i.e. management intensity), our study necessarily left out other types of response, such as resource selection, species interactions or levels of philopatry, that are inherent to single species. However, changes in bird species richness and/or abundance are linked to changes in available resources and structures (MacArthur and MacArthur, 1961). Therefore, changes in the numerical response can constitute a first alarm bell, which led to the establishment of monitoring schemes based on these metrics (Gregory and van Strien, 2010).

The results highlight that several confounding factors affect the mitigation effect that retention forestry can provide on birds. Species' natural history, the surrounding environment, past management or disturbance regime as well as current management, can all drive species abundance and richness and foster reactions in the assemblages (Balestrieri et al., 2015;

Bergeron et al., 2017; Jetz et al., 2012; Korňan and Adamík, 2017; Ram et al., 2017). Our findings allow us to infer general levels of retention needed to keep the abundance of particular avian guilds at levels comparable with unharvested forests. In our study, retention level alone was apparently not sufficient for explaining numerical responses of bird assemblages. Canopy and cavity nesters and bark foragers showed a more positive response in abundance to lower levels of retention in mixed or broadleaved forests than in conifer forests. However, the response of canopy nesters in abundance was influenced by latitude as well. Our results show that at temperate latitude, retention levels can have a null or slightly negative influence on their abundance, contrasting with the pattern found at boreal latitudes. The difference can be the consequence of the past and current management regimes: a simplified forest landscape at boreal latitudes (Imbeau et al., 2001), where leaving more trees after logging increases the canopy volume, compared to the main management regime, i.e. clearcutting (e.g. Söderström, 2009) or, in temperate biomes, a forest landscape mostly relegated to higher elevation in temperate latitudes or extremely fragmented, which rarely reflects the original bird assemblage. Bark foragers also presented contrasting patterns between forest types, with strong positive effects of retention levels in mixed forests. Bark foragers can benefit not only by the simple increase of volume of feeding substrates (e.g. Tremblay et al., 2015), namely tree stems, but also from a diversity of tree species, more similar to natural forests, which are related to a greater diversity in insect assemblages (Pettersson et al., 1995). This is also in line with earlier findings showing high importance of local factors on bark foragers in temperate forests in Oregon (Galitsky and Lawler, 2015). Foliage forager abundance, however, showed a different pattern, with broadleaved and conifer forests hosting higher numbers of birds. This pattern was more pronounced in temperate than in boreal forests. Species richness, instead, showed a different pattern than abundance. For some guilds, such as canopy nesters and ground foragers, retention level positively influenced species richness. But other factors were also important. For example, cavity nesters richness showed different responses in boreal and temperate forests, with null or limited responses in broadleaved and conifer forests, and a stronger response in mixed forests. In particular, the negative response recorded in temperate forests may be a signal that in more diverse forest landscapes, such as at temperate latitudes, retention levels alone cannot explain the variability we observe in the bird assemblage, and that other variables, not covered in our analysis, can play a major role as well. For example, Lencinas et al. (2018), in a study in Southern Patagonia, found high importance of proximity of forests edges for bird species occurring in forest landscapes being subject to retention forestry. Also, a diversified pattern of species response to different types of forest management was recently reported by Kellner et al. (2016) and Perry et al. (2018).

4.2. Identifying bias and knowledge gaps in the literature

In our summary of the state-of-the-art and knowledge-based evidence on the effects of retention levels on birds, we found two main knowledge domains, which can be improved by further research: theoretical and methodological.

From a theoretical point of view, several issues can affect the results. In this study, we focused only on the numerical response of birds to management, but it is worth mentioning that management strongly affects the functional response of birds, as well (e.g. Balestrieri et al., 2018; Suorsa et al., 2004). This effect can potentially introduce uncertainty in our conclusion. The effects of disturbance on birds, for instance, can be less detectable within large areas of undisturbed habitat (Schmiegelow et al., 1997). Also, fragmentation can be a major driver of species distribution (Basile et al., 2016; Devictor et al., 2008), although overall species richness has potential to benefit from it (Fahrig, 2017). Our findings about bird assemblage responses to cutblock size are concordant with theory, highlighting a bell-shaped response to patch size for many guilds and suggesting that the amount of habitat and the surrounding matrix are strong determinants of the bird assemblage composition (e.g. Brotons et al., 2003). The findings also highlight the importance of the matrix in which the studies occurred due to spill-over effects (Tscharntke et al., 2012). The effect of forest patch area on bird abundance in harvested landscapes appears generally to be less pronounced than in the case of true forest islands (Brotons et al., 2003b). The observed decrease in effect size for larger patch sizes suggests that the spill-over effect can be diluted over large areas, resulting in lower abundances when compared to large unharvested forests. Large patch aggregation can mitigate edge effects, decreasing the number of individuals (Mori et al., 2017). Therefore, we look at two processes working in different directions. Given that most of the studies deployed study areas 10–30 ha large, we identify the lack of small- and large-scale studies as a flaw.

Another source of variability is post-disturbance conditions (Balestrieri et al., 2015; Tozer et al., 2010), as birds can still show a high degree of resilience after ~10 years (Bergeron et al., 2017). We found most of the studies deployed their experimental design over a short time-scale, raising questions about the reliability of the results. As a matter of fact, species abundance curves of tree-dependent guilds show that guilds can recover from disturbance after 10 + years, probably in response of forest regeneration (Millington et al., 2011). On the other hand, ground-dependent species show a decrease over the years, given the different light conditions of retention cuts. Therefore, we identified the lack of long-term studies as the second gap researchers should fill in.

Methodological biases arose as our criteria automatically excluded most European articles and many others because of insufficient data reporting. In such cases, we believe it is up to journal editors and reviewers to ask the authors to include important ancillary data in their articles, such as full species inventories, or descriptive statistics of the most used forest variables, such as mean dbh, tree density, species composition and so on (Vetter et al., 2015). Fortunately, the recent trend, now being adopted by an increasing number of journals, is to require full datasets, and providing more space in 'supplementary information' can improve the situation.

Finally, the last issue concerns the unbalanced experimental design *i.e.* availability of different treatments, which, in most cases, is not manageable by researchers. Biodiversity research in forests is usually constrained by forest management systems, leaving researchers very little flexibility, and forcing them to adapt the study design to the system.

4.3. Ecological evidence and management implications

The concept of retention forestry is ecologically grounded in the theory of disturbance (Picket and White, 1985). Multiple-use forests, among other objectives, seek to preserve biodiversity components and functionality by conserving legacies which act as “lifeboats” for biodiversity, in the same way as such structures are left in place after natural disturbances (e.g. fires, wind-felling, insect outbreaks) (Johnstone et al., 2016) and to which species are evolutionary adapted (Drapeau et al., 2016). In modified landscapes, the amount and distribution of different forests types, succession phases and structures is highly different from the original (Bengtsson et al., 2003). The current levels of retention are rarely above 10%, with some North American exceptions going up to 20% (Gustafsson et al., 2012). Therefore, even if we assume that retention patches are able to mimic unharvested forests in several dimensions, they are not able to encapsulate much of the variation observed in naturally dynamic forest landscapes. In line with that, our approach, by simulating a model landscapes, required much higher retention levels. Guilds like cavity nesters and aerial foragers showed that their abundance in harvested forests is still lower than in unharvested forests, even when the retention level is above 20%. Canopy nesters and foliage foragers, instead, showed that retention levels of 20% might be enough in temperate or boreal forests, respectively, to preserve their numbers as in unharvested forests. However, at such a retention level, canopy nesters show decreased richness. Our approach showed that globally meaningful retention levels are impossible to find, as each guild shows different responses to the practice. Instead, the levels should be evaluated by taking into account the array of species targeted for conservation. Direct application of higher levels in practice would result in a loss in timber production, which could be compensated only by increasing the harvested area and management intensity elsewhere. Moreover, retention forestry is today mostly applied in areas <20 ha (Gustafsson et al., 2012), while our findings show that positive effects could be still found in 30 + ha patches. In our view, current practices and the ecological responses of birds are not aligned. To fully embody their role as ‘lifeboats’, retention forestry systems might need up to a couple of centuries, as showed by simulations (Seidl et al., 2014). *Ad hoc* retention, by choosing specific trees, can speed up the process, as such trees, usually larger and older ones, can support more structural diversity (Paillet et al., 2017). Large trees, for instance, have a higher probability to remain in the forest in case of disturbance (Johnstone et al., 2016), making retention forestry more similar to natural events than clearcutting, which only presents very superficial similarities with natural disturbances (Bengtsson et al., 2000).

On the other hand, restoration practitioners can view our results as a tool for effective planning. In many regions of the world, there is an ongoing process of reforestation with increasing interest in re-establishing a more natural forest landscape (Stanturf et al., 2014). The retention levels we found to be necessary for maintaining the entire forest bird assemblage show that specific targets for restoration in a landscape context can be set. Bird status in restored forest can be enhanced up to 84%, compared to degraded forests (Crouzeilles et al., 2016).

The findings of our study are also relevant in the context of the world's intact forests having exceptional value for biodiversity (Watson et al., 2018). Unfortunately, such intact or little-impacted forests have been continuously logged both in developing and developed countries as the forest logging frontier moves to remote areas (Potapov et al., 2017; Svensson et al.,

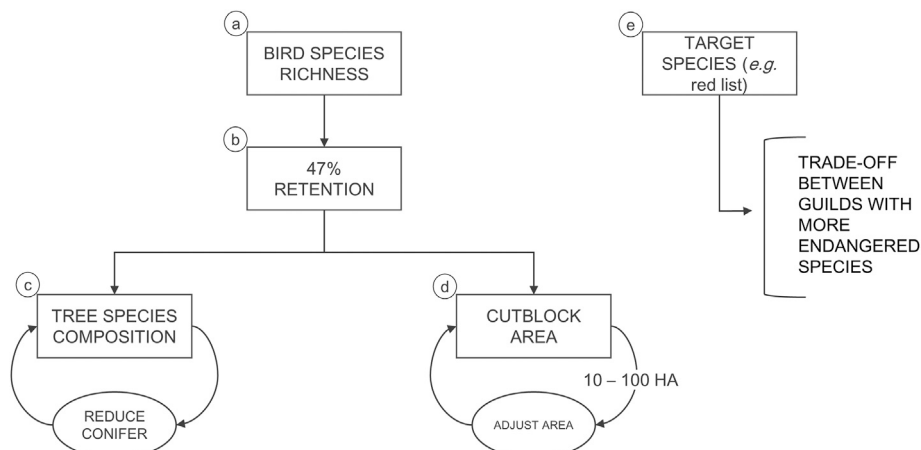


Fig. 4. Management suggestions to cope with different bird conservation target and help improving retention harvests. Depending on the possibility to achieve the expected retention threshold, one can manage different forest features. If the aim is to increase bird species richness (a) and the 47% level is not achievable (b), then managers should opt for increasing tree species richness (c) and management unit area (d). If the aim is target species (e) then managers need to look at species-specific levels.

2019). Our results, based on the analysis of forest bird assemblages, suggests minimum tree retention levels should be set at 40–60% if the goal is to preserve the entire bird assemblage found in unharvested forest landscapes.

Landscape heterogeneity and stand structural complexity have been proposed as driving principles for forest biodiversity conservation (Lindenmayer et al., 2006). From a practitioner's point of view, this is translated in planning the size and timing of harvesting units, as well as the key structures to focus on. We exemplified an application of our findings in a flowchart to help forest management fulfil the conservation of specific bird assemblages (Fig. 4). According to our study, maintaining natural levels of bird diversity indeed requires a very high retention level. Though this might seem unfeasible in multiple-use forests, conservation targets require strong actions. Indeed, we remark that retention levels are one of the possible targets to work on, but tree species composition, size of the cutblock and rotation time should also be considered when planning harvest operations. Our current understanding of bird assemblages derives mostly from already impoverished communities (including in the majority of old forests serving as reference in this study), from which the specialised share of species was already lost. Therefore, our current conservation targets might underestimate actual biodiversity potential, and might be insufficient to fully conserve or restore original communities. A compromise could lie in increasing the mix of native tree species, including both conifers and broadleaved, over larger patches, ranging 10–100 ha.

Finally, it is worth remarking that retention practices can actually benefit a larger portion of biodiversity than just birds. Our recommendations are meant to mitigate habitat degradation and isolation, which can be major drivers of habitat specialist decrease (Fischer and Lindenmayer, 2007).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gecco.2019.e00615>.

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